

Black body radiation

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know.

QUESTION		ANSWER	✓
1	Name the type of radiation that all objects emit.	Infrared radiation	☐
2	Describe how the infrared radiation emitted by an object depends on its temperature.	The hotter the object, the more it emits.	☐
3	What is a perfect black body?	An object that absorbs all the radiation that falls on it.	☐
4	State how an object's ability to absorb radiation relates to its ability to emit radiation.	A good absorber is also a good emitter.	☐
5	Describe how the radiation emitted by an object changes as its temperature increases.	The intensity increases and the peak moves to a shorter wavelength.	☐
6	An object is at a constant temperature. Compare the radiation it emits and absorbs.	It absorbs and emits radiation at the same rate.	☐
7	State when the temperature of an object increases, in terms of radiation.	When it absorbs radiation faster than it emits it.	☐
8	State what all objects do, whatever their temperature.	They emit and absorb infrared radiation.	☐
9	State when the temperature of an object decreases, in terms of radiation.	When it emits radiation faster than it absorbs it.	☐
10	Give three factors that the temperature of the Earth depends on.	The rates at which it absorbs and emits radiation, and how much is reflected into space.	☐